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Falling Interest Rates and Credit Reallocation: Lessons from General Equilibrium

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1 Big picture

- **Question:** In an economy with heterogeneous entrepreneurs and borrowing constraints, does a fall in the interest rate always stimulate aggregate activity, or can it instead worsen the allocation of capital and reduce output?
- **Setting:** A canonical small-open-economy model with entrepreneurs that differ in productivity, face financial frictions, and demand a factor whose supply is not perfectly elastic. The paper later interprets this factor as land or real estate and embeds the mechanism in a dynamic macro-finance environment.
- **Core empirical object:** The paper studies how lower interest rates affect the allocation of capital across low- and high-productivity firms, not just the aggregate amount of capital.
- **Main theoretical result:** In partial equilibrium, lower rates relax borrowing constraints and induce more firms to invest. In general equilibrium, however, they also raise the price of capital and crowd out investment by more productive constrained firms. If this reallocation is strong enough, lower rates reduce aggregate output.
- **Main dynamic result:** In the short run, lower rates can generate a boom through balance-sheet effects, because higher asset prices raise the wealth of productive borrowers. But that channel is temporary. If rates remain low, the negative reallocation effect dominates and output eventually falls.
- **Main quantitative result:** Calibrated to U.S. data and the long decline in the real rate since the mid-1980s, the model implies a long-run output decline of about 5.2%.
- **Main empirical result:** In U.S. firm data, lower rates have a stronger investment effect on low-MPK firms than on high-MPK firms exactly where local general-equilibrium effects are stronger.
- **Why we care:** The paper shows that low rates do not only shift aggregate demand. They can also distort the allocation of scarce capital. That is the novel contribution.

2 Data and measurement (key objects)

2.1 How the paper interprets “capital”

- **Theory object:** Any factor of production whose price depends on local demand and supply, so that firm investment decisions feed back into equilibrium prices.
- **Preferred interpretation:** Land or real estate. This makes capital-supply elasticity empirically meaningful and naturally links interest rates to asset prices, collateral values, and local crowding-out effects.
- **Broader interpretation:** The mechanism would also apply to other scarce, credit-intensive inputs, such as skilled labor financed through working capital.

2.2 Calibration inputs for the dynamic model

- **Interest-rate path:** The paper feeds into the model the smoothed long-run real interest rate estimated by Bauer and Rudebusch (2020), declining from roughly 1.97% in 1985:Q3 to roughly 0.62% in 2018:Q3.
- **Productivity process:** Entrepreneurial productivity follows a Poisson-driven stochastic process calibrated to match firm-level persistence and volatility, with $\theta = 1$, $\gamma_A = 0.8$, and $\sigma_A = 0.45$.
- **Financial friction:** The borrowing limit is set to $\lambda = 0.65$, matching non-financial firms' debt-to-asset ratios.
- **Capital supply:** The elasticity of capital supply is $\varepsilon = 2$, motivated by the average U.S. real-estate supply elasticity in Saiz (2010).
- **Other parameters:** Depreciation is $\delta = 6.5\%$ and the subjective discount rate is $\rho = 4\%$.

2.3 Data for the empirical exercise

- **Firm data:** Quarterly Compustat data for U.S. publicly listed firms, 1990:Q1–2019:Q2.
- **Outcome variable:** Investment is $\Delta \log k_{i,t+1}$, the log change in the firm's real tangible capital stock.
- **Regional asset-price measure:** Quarterly real house-price growth by MSA from FHFA.
- **Source of cross-sectional pass-through:** MSA-level real-estate supply elasticities from Saiz (2010).
- **Interest-rate measure:** The baseline uses the real 30-year mortgage rate from Freddie Mac. The appendix also uses high-frequency monetary shocks from Jarocinski and Karadi (2020).
- **Firm productivity proxy:** Firm-level MPK, constructed as the output elasticity of capital times real value added divided by the real capital stock. Sector elasticities are estimated from production functions using Wooldridge (2009).
- **Controls:** MPK, sales growth, size, current assets / total assets, debt / total assets, and MPK interacted with lagged real GDP growth.
- **Caveat:** Since Compustat includes only listed firms, the measured importance of financing frictions is likely conservative.

3 Models and empirical specifications (all core equations + interpretation)

3.1 Baseline static model

Entrepreneurs choose borrowing b_A and capital k_A to maximize expected consumption:

$$\max_{b_A, k_A} Ak_A - Rb_A \quad (1)$$

$$\text{s.t. } qk_A \leq w + b_A, \quad (2)$$

$$Rb_A \leq \lambda Ak_A, \quad (3)$$

$$k_A \geq 0. \quad (4)$$

- **Interpretation:** Entrepreneurs compare the return on capital, A/q , with the world interest rate R , but they cannot fully borrow against future output because only a fraction λ is pledgeable.
- **Economic intuition:** Financially constrained but productive entrepreneurs want to lever up and buy capital; the friction limits how much they can do so.

Optimal capital demand is:

$$k_A(q, R) = \begin{cases} 0, & \text{if } \frac{A}{q} < R, \\ \in \left[0, \frac{w}{q - \lambda A/R}\right], & \text{if } R = \frac{A}{q}, \\ \frac{w}{q - \lambda A/R}, & \text{if } \frac{\lambda A}{q} < R < \frac{A}{q}, \\ \infty, & \text{if } R \leq \frac{\lambda A}{q}. \end{cases} \quad (5)$$

Borrowing satisfies:

$$b_A(q, R) = qk_A(q, R) - w. \quad (6)$$

- **Infra-marginal firms:** If $A/q < R$, they prefer to save rather than invest.
- **Marginal firm:** If $A = qR$, the firm is indifferent.
- **Supra-marginal firms:** If $A/q > R$, they invest; when $R > \lambda A/q$, their borrowing constraint binds.

Aggregate demand, supply, market clearing, and output are:

$$K^D(q, R) \equiv \int_0^1 k_A(q, R) dG(A), \quad (7)$$

$$K^S(q) \in \arg \max_{k \geq 0} \{qk - \chi(k)\}, \quad (8)$$

$$K^S(q) = K^D(q, R), \quad (9)$$

$$Y(q, R) = \int_0^1 Ak_A(q, R) dG(A). \quad (10)$$

- **Interpretation:** Output depends both on the total amount of capital and on who uses that capital. This is exactly where misallocation enters.

3.2 How lower interest rates affect investment and output

For a supra-marginal entrepreneur,

$$\frac{dk_A}{dR} = \frac{\left| \frac{dq}{dR} \right| - \frac{\lambda A}{R^2}}{q - \frac{\lambda A}{R}} k_A. \quad (11)$$

- **Direct or partial-equilibrium effect:** The term $\lambda A/R^2$ captures the standard force: lower R raises the pledgeable value of output and relaxes borrowing constraints.
- **General-equilibrium effect:** The term involving dq/dR captures the increase in the price of capital caused by stronger investment demand. That crowds out constrained investors.
- **Key implication:** Lower rates always draw in some lower-productivity firms, but may reduce investment by some higher-productivity firms if q responds strongly enough.

The output effect decomposes into:

$$\frac{dY}{dR} = \underbrace{\int_{qR}^1 (A - qR) \frac{dk_A}{dR} dG(A)}_{\text{capital-reallocation effect } \mathcal{R}} + \underbrace{qR \frac{dK^S(q)}{dR}}_{\text{capital-supply effect } \mathcal{K}}. \quad (12)$$

- **Interpretation of $\mathcal{R}$:** This measures how output changes because capital is reallocated across investors.
- **Interpretation of $\mathcal{K}$:** This is the standard extensive-margin effect: lower rates raise demand and, if supply is elastic, the capital stock expands.
- **Sign logic:** $\mathcal{K} \leq 0$ because lower rates increase the capital stock. The sign of $\mathcal{R}$ is ambiguous; it is positive when lower rates are contractionary through misallocation.
- **Proposition 1:** Lower capital-supply elasticity ε strengthens $\mathcal{R}$, weakens $\mathcal{K}$, and raises dY/dR . For sufficiently low ε and sufficiently tight borrowing constraints, $dY/dR > 0$, so a fall in R lowers output.

3.3 Robustness: diminishing returns and heterogeneous endowments

With technology $Af(k)$ and heterogeneous wealth w , the first-best scale $k_A^*(q, R)$ satisfies $Af'(k_A^*)/q = R$. Optimal investment is:

$$k_{(A,w)}(q, R) = \begin{cases} k_A^*(q, R), & \text{if } qk_A^*(q, R) \leq w + \frac{\lambda Af(k_A^*(q, R))}{R}, \\ k \text{ such that } qk = w + \frac{\lambda Af(k)}{R}, & \text{otherwise.} \end{cases} \quad (13)$$

Market clearing and output become:

$$K^S(q) = \int k_{(A,w)}(q, R) dG(A, w), \quad (14)$$

$$Y = \int Af(k_{(A,w)}(q, R)) dG(A, w). \quad (15)$$

The output effect is:

$$\frac{dY}{dR} = \int_{(A,w) \in C} \left(Af'(k_{(A,w)}(q, R)) - qR \right) \frac{dk_{(A,w)}(q, R)}{dR} dG(A, w) + qR \frac{dK^S(q)}{dR}, \quad (16)$$

where C is the set of constrained entrepreneurs.

- **Interpretation:** The reallocation mechanism does not rely on linear technology. What matters is heterogeneous MPK, not just heterogeneous average productivity.
- **Takeaway:** Lower rates can shift capital from high-MPK, constrained firms toward low-MPK, less constrained firms even when some large productive firms are unconstrained.

3.4 Normative benchmark: constrained planner

The planner solves:

$$\max_{\{k_A\}} \int_0^1 Ak_A dG(A) - R[\chi(K^S) - w], \quad (17)$$

$$\text{s.t. } R(qk_A - w) \leq \lambda Ak_A, \quad k_A \geq 0 \forall A, \quad (18)$$

$$\chi'(q) = K^S = \int_0^1 k_A dG(A). \quad (19)$$

The social net present value of one more unit of investment by entrepreneur A is:

$$\text{NPV}_A^{\text{SP}} \equiv \frac{A}{R} - q - \chi''(K^S) \left(\int \gamma_{\tilde{A}} k_{\tilde{A}} dG(\tilde{A}) \right), \quad (20)$$

where γ_A is the multiplier on the entrepreneur's borrowing constraint.

- **Interpretation:** Individual firms ignore that their extra demand raises q , which tightens borrowing constraints for other constrained firms. This is a pecuniary externality with first-order allocative consequences.
- **Planner implication:** The planner sets a higher productivity threshold than the market, $\tilde{A} > q^{\text{CE}} R$, and excludes some low-productivity investors to free up capital for more productive users.
- **Proposition 2:** In the constrained-efficient allocation, $dY^{\text{SP}}/dR < 0$. So lower rates are always expansionary once the externality is internalized.
- **Economic intuition:** The competitive problem is not that lower rates are inherently bad. It is that the market reallocates scarce capital in the wrong direction.

3.5 Dynamic model with wealth accumulation

Entrepreneurial wealth evolves according to:

$$\dot{w} = y + \dot{q}k - \delta qk - rb - c. \quad (21)$$

The collateral constraint is:

$$b \leq \lambda qk. \quad (22)$$

Capital accumulation satisfies:

$$\dot{K} = I(q) - \delta K, \quad (23)$$

$$I(q) \equiv \arg \max_{I \geq 0} \{qI - \chi(I)\}. \quad (24)$$

Optimal policies are:

$$k = \begin{cases} \frac{1}{1-\lambda} \frac{w}{q}, & \text{if } A + \dot{q} \geq (r + \delta)q, \\ 0, & \text{otherwise,} \end{cases} \quad (25)$$

$$b = qk - w, \quad (26)$$

$$c = \rho w. \quad (27)$$

Wealth dynamics become:

$$\dot{w} = \begin{cases} \left(\left[\frac{A + \dot{q}}{q} - \delta - \lambda r \right] \frac{1}{1-\lambda} - \rho \right) w, & \text{if } A + \dot{q} \geq (r + \delta)q, \\ (r - \rho)w, & \text{otherwise.} \end{cases} \quad (28)$$

- **Interpretation:** Wealth now responds endogenously to productivity, interest rates, and capital gains. Productive firms accumulate faster; borrowers gain from lower rates; savers lose.

Aggregate wealth by productivity type is:

$$W_A \equiv \int w f(A, w) dw, \quad (29)$$

$$W \equiv \int W_A dA. \quad (30)$$

Its law of motion is:

$$\dot{W}_A = \begin{cases} \left(\left[\frac{A + \dot{q}}{q} - \delta - \lambda r \right] \frac{1}{1-\lambda} - \rho - \theta \right) W_A + \theta \int g(A | A') W_{A'} dA', & \text{if } A + \dot{q} \geq (r + \delta)q, \\ (r - \rho - \theta) W_A + \theta \int g(A | A') W_{A'} dA', & \text{otherwise.} \end{cases} \quad (31)$$

Per-capita investment, capital-market clearing, and output are:

$$k_A = \begin{cases} \frac{1}{1-\lambda} \frac{W_A}{q g(A)}, & \text{if } A + \dot{q} \geq (r + \delta)q, \\ 0, & \text{otherwise,} \end{cases} \quad (32)$$

$$K = \int_{A \geq (r+\delta)q - \dot{q}} k_A g(A) dA, \quad (33)$$

$$Y = \int_{A \geq (r+\delta)q - \dot{q}} A k_A g(A) dA. \quad (34)$$

- **Threshold condition:** The dynamic marginal entrepreneur is defined by $A + \dot{q} = (r + \delta)q$.
- **Key difference from the static model:** The reallocation effect now interacts with endogenous wealth accumulation and with capital gains.

3.6 Steady states and transition dynamics

In the analytically tractable case with fixed capital supply and productivity redraws independent of current productivity,

$$\frac{W_A}{W} = \begin{cases} \frac{\theta}{\theta + \rho - \frac{1}{1-\lambda} \left(\frac{A}{q} - \lambda r \right)} g(A), & \text{if } A \geq rq, \\ \frac{\theta}{\theta + \rho - r} g(A), & \text{otherwise.} \end{cases} \quad (35)$$

Wealth shares must sum to one:

$$\int_{rq}^1 \frac{\theta}{\theta + \rho - \frac{1}{1-\lambda} \left(\frac{A}{q} - \lambda r \right)} g(A) dA = 1 - \int_0^{rq} \frac{\theta}{\theta + \rho - r} g(A) dA. \quad (36)$$

Capital-market clearing is:

$$\bar{K} = \left[\int_{rq}^1 \frac{\theta}{\theta + \rho - \frac{1}{1-\lambda} \left(\frac{A}{q} - \lambda r \right)} g(A) dA \right] \cdot \frac{1}{1-\lambda} \frac{W}{q}. \quad (37)$$

The steady-state output effect is:

$$\frac{dY}{dr} = \int_{rq}^1 (A - rq) \frac{dk_A}{dr} g(A) dA. \quad (38)$$

- **Interpretation of (35)–(37):** In steady state, lower rates transfer wealth from creditor-type entrepreneurs to debtor-type entrepreneurs and reduce the productivity threshold rq .
- **Interpretation of (38):** The same reallocation logic from the static model survives in steady state: lower rates can shift capital from high- to low-productivity users.

When an unanticipated rate shock hits at t_0 , wealth jumps to:

$$W_{A,t_0^+} = \begin{cases} \left(\frac{q_{t_0^+}}{q_{t_0}} - \lambda \right) \frac{1}{1-\lambda} W_{A,t_0}, & \text{if } A \geq (r_{t_0} + \delta) q_{t_0}, \\ W_{A,t_0}, & \text{otherwise.} \end{cases} \quad (39)$$

and the impact jump in the capital price is pinned down by:

$$\frac{q_{t_0^+}}{q_{t_0}} = \lambda \cdot \frac{\int_{A \geq r} q_{t_0^+} - \dot{q}_{t_0^+} W_{A,t_0} dA}{\int_{A \geq r} q_{t_0^+} - \dot{q}_{t_0^+} W_{A,t_0} dA - (1-\lambda) \int_{A \geq r q_{t_0}} W_{A,t_0} dA}. \quad (40)$$

- **Interpretation:** Lower rates raise the price of capital on impact and therefore the wealth of the firms that already own capital. This is the balance-sheet channel.
- **Crucial distinction:** Balance-sheet effects come from unexpected asset-price changes, so they are temporary. The reallocation effect lasts for as long as rates remain low.

3.7 Quantitative decomposition

For the numerical exercise with a discrete interest-rate change, the paper decomposes output as:

$$\frac{Y_t - Y_0}{Y_0} = \underbrace{\frac{\int_{\hat{A}_t}^{\hat{A}_0} (A - \bar{A}_t)(k_{A,t} - k_{A,0})g(A) dA}{Y_0}}_{\mathcal{R}_t} + \underbrace{\bar{A}_t \frac{K_t - K_0}{Y_0}}_{\mathcal{K}_t}, \quad \hat{A}_t = (r_t + \delta)q_t - \dot{q}_t. \quad (41)$$

- **Interpretation:** The paper separately measures how much of the output response comes from changes in the allocation of capital across firms versus changes in the aggregate capital stock.

3.8 Empirical specifications

The baseline panel regression is:

$$\Delta \log k_{i,t+1} = \alpha_i + \alpha_{jt} + \beta \Delta \log q_{kt} + \gamma \Delta r_t + \varepsilon_{it}. \quad (42)$$

The first-stage instrument for local asset-price growth is:

$$\Delta \log q_{kt} = \alpha_k + \alpha_t + \rho H_k \Delta r_t + \varepsilon_{kt}, \quad (43)$$

where H_k is local real-estate supply elasticity.

The key heterogeneous-effects regression is:

$$\Delta \log k_{i,t+1} = \alpha_i + \alpha_{jt} + \beta_1 \Delta \log q_{kt} + \gamma_1 \Delta r_t + \beta_2 \text{mpk}_{i,t-1} \Delta \log q_{kt} + \gamma_2 \text{mpk}_{i,t-1} \Delta r_t + \Gamma' Z_{i,t-1} + \varepsilon_{it}. \quad (44)$$

The local-projection version is:

$$\log k_{i,t+h} - \log k_{i,t-1} = \alpha_{ih} + \alpha_{jth} + \beta_{1h} \Delta \log q_{kt} + \gamma_{1h} \Delta r_t + \beta_{2h} \text{mpk}_{i,t-1} \Delta \log q_{kt} + \gamma_{2h} \text{mpk}_{i,t-1} \Delta r_t + \Gamma'_h Z_{i,t-1} + \varepsilon_{ith}. \quad (45)$$

- **Dependent variable:** Firm investment.
- **Regressor of interest:** The interaction of MPK with real-estate price growth, β_2 , which captures whether lower-MPK firms benefit more from the general-equilibrium asset-price response.
- **Main prediction:** $\beta_2 < 0$. Higher real-estate prices induced by lower rates should raise the relative investment of lower-MPK firms and crowd out higher-MPK firms.
- **Direct effect versus GE effect:** γ_2 measures whether high-MPK and low-MPK firms respond differently to the direct change in rates holding local asset prices fixed.

4 Identification and instruments (what makes the estimates credible)

4.1 Theory-side identification

- **Identifying variation:** In the main theory, the economy is small and open, so the interest rate is taken as exogenous.

- **What is being compared:** The paper compares the partial-equilibrium response to lower rates with the full general-equilibrium response when the capital price adjusts.
- **Main concern:** If interest rates were fully endogenous to domestic conditions, one might worry that the mechanism is an artifact of the small-open-economy assumption.
- **Response:** The online appendix shows the same logic in a closed economy where lower rates emerge from a savings glut. So the reallocation channel is not tied to exogenous world-rate movements.

4.2 Empirical identification

- **Baseline challenge:** Changes in local real-estate prices are endogenous. They may reflect local demand shocks that also affect firm investment.
- **Instrument:** The paper instruments local real-estate price growth with the interaction between the common interest-rate change and local real-estate supply elasticity, $H_k \Delta r_t$.
- **Relevance:** The first stage is strong. Real-estate prices react more to common rate changes in locations where supply is less able to adjust.
- **Exclusion restriction:** Conditional on firm fixed effects and sector-time fixed effects, the interaction should affect firm investment mainly through local real-estate prices rather than through independent firm-level shocks.
- **Fixed effects:** Firm fixed effects absorb time-invariant firm heterogeneity. Sector-time fixed effects absorb common industry-level shocks and the common component of interest-rate movements.
- **Why the interaction with MPK matters:** The theory is specifically about differential reallocation across firms. The empirical test therefore asks whether asset-price changes induced by lower rates disproportionately shift investment toward low-MPK firms.
- **Inference:** The paper reports two-way clustered standard errors by firm and quarter, and bootstrapped standard errors for the 2SLS specifications.
- **Robustness:** Results are similar when the paper replaces mortgage-rate variation with high-frequency monetary-policy surprises from Jarocinski and Karadi (2020).

5 Tables and figures

5.1 Figure 1: Distribution of capital across entrepreneurs given prices $\{q, R\}$

Read:

- The figure draws the static capital-demand schedule across entrepreneur productivity.
- Firms below the threshold $A = qR$ do not invest; firms above it invest and are constrained.
- The visual is helpful because it makes clear that the equilibrium is organized around a productivity threshold, and that investment by constrained firms is increasing in productivity.

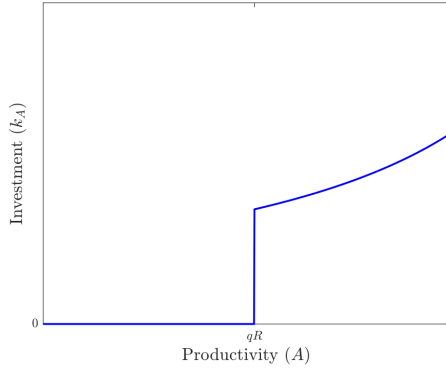


FIGURE 1
Distribution of capital across entrepreneurs given prices (q, R)

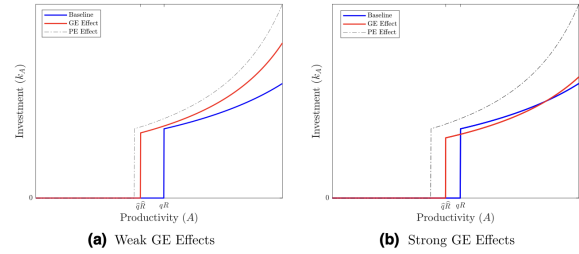


FIGURE 2
Partial- and general-equilibrium effects of a fall in the interest rate

5.2 Figure 2: Partial- and general-equilibrium effects of a fall in the interest rate

Read:

- Panel (a) shows a case with weak general-equilibrium effects: lower rates expand investment for all active firms.
- Panel (b) shows strong general-equilibrium effects: the rise in the capital price is so strong that some high-productivity firms reduce investment even though rates fall.
- This is the core picture of the paper. Lower rates do not simply “shift demand out.” They also move the capital price enough to change who gets crowded out.

5.3 Figure 3: Effect of the elasticity of capital supply on the response of output

Read:

- The top curve logic is that lower supply elasticity makes the reallocation effect more contractionary.
- The capital-supply effect becomes weaker when supply is inelastic, because the aggregate capital stock adjusts less.
- With tight enough constraints, the total output derivative crosses zero: that is the region where lower rates become contractionary.

5.4 Figure 4: A permanent fall in the interest rate

Read:

- The left panel shows a permanent drop in the rate.
- The middle panel shows a jump and hump in the asset price. With balance-sheet effects, asset prices rise more and stay elevated for some time.
- The right panel shows a boom-bust response of output. Output initially rises because the balance-sheet channel helps productive borrowers, but later falls below baseline once the persistent

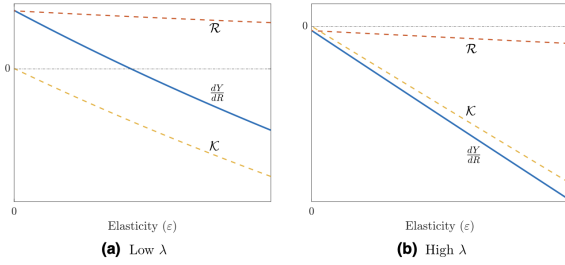


FIGURE 3
Effect of the elasticity of capital supply on the response of output to a change in the interest rate

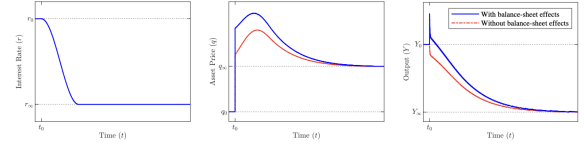


FIGURE 4
A permanent fall in the interest rate

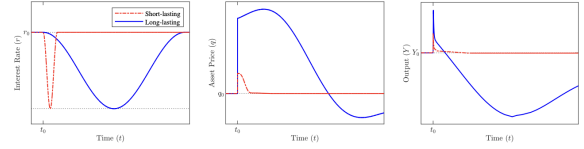


FIGURE 5
A temporary fall in the interest rate

reallocation effect dominates.

5.5 Figure 5: A temporary fall in the interest rate

Read:

- The figure compares a short-lived and a long-lived temporary rate decline.
- When the shock is very short-lived, balance-sheet effects dominate and output remains positive.
- When the shock is more persistent, the economy still booms at first but later experiences a bust as the reallocation effect outlasts the balance-sheet effect.

5.6 Table 1: Calibration values

Read:

- The key calibrated ingredients are moderate leverage ($\lambda = 0.65$) and a capital-supply elasticity of 2.
- Productivity is persistent but not perfectly so ($\gamma_A = 0.8$), and volatile enough ($\sigma_A = 0.45$) to generate meaningful heterogeneity.
- These values discipline the quantitative exercise and clarify which assumptions are doing most of the work.

TABLE 1 The parameter values used for the numerical exercise		
Parameter	Value	Source
Frequency of transition of log productivity	$\theta = 1$	Yearly frequency
Persistence of disturbance to log productivity	$\gamma_A = 0.8$	Foster <i>et al.</i> (2008)
Volatility of disturbance to log productivity	$\sigma_A = 0.45$	Bloom <i>et al.</i> (2018)
Borrowing limit (ratio of debt to assets)	$\lambda = 0.65$	Kalemli-Ozcan <i>et al.</i> (2012)
Elasticity of capital-supply	$\varepsilon = 2$	Saiz (2010)
Depreciation rate of capital	$\delta = 6.5\%$	Khan and Thomas (2008)
Subjective time discount rate	$\rho = 4\%$	Lustig <i>et al.</i> (2013)

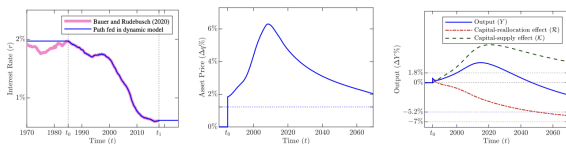


FIGURE 6
Quantitative effects from declining real interest rates

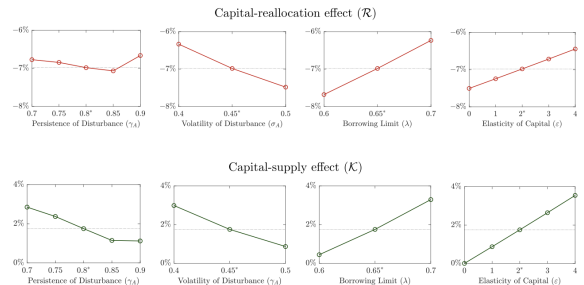


FIGURE 7
Sensitivity analysis

5.7 Figure 6: Quantitative effects from declining real interest rates

Read:

- The left panel feeds in the long-run decline in the real rate.
- The middle panel shows a sizeable asset-price increase, peaking around the transition and then gradually declining.
- The right panel decomposes output. The capital-supply effect is positive, but the reallocation effect becomes strongly negative and dominates in the long run, leading to an overall output decline of about 5.2%.

5.8 Figure 7: Sensitivity analysis

Read:

- More severe financial frictions strengthen the contractionary reallocation effect substantially.
- Lower capital-supply elasticity also makes reallocation more contractionary and dampens the capital-supply offset.
- Productivity volatility matters more than productivity persistence for the size of the reallocation effect.

5.9 Table 2: Investment response to real-estate prices and interest rates

Read:

- The average effect of local real-estate price growth on investment is positive in OLS, but not precisely estimated in 2SLS.
- The key interaction, $MPK \times$ real-estate price growth, is negative and statistically significant in both OLS and 2SLS. In the main 2SLS specification the coefficient is about -0.127 , and it remains about -0.095 with the full control set.
- The interaction $MPK \times$ interest-rate change is small and insignificant, which supports the paper's interpretation that the differential response works mainly through the general-equilibrium asset-price channel, not through a direct rate channel.
- Economically, a 1.8 percentage point increase in real-estate prices raises investment by about 0.23 percentage points more for firms one standard deviation lower in MPK, which is large relative to the sample average investment rate of 0.5%.

5.10 Figure 8: Dynamic differential response of investment to real-estate price changes

Read:

- The local-projection coefficients become progressively more negative over the first year after the shock.

TABLE 2
Investment response to real estate (RE) prices and interest rates

	(1)	(2)	(3)	(4)	(5)
RE price growth	0.078*** (0.023)	0.066 (0.876)	0.058** (0.023)	-0.145 (0.884)	-0.195 (0.800)
MPK x RE price growth			-0.053*** (0.017)	-0.127*** (0.039)	-0.095*** (0.037)
MPK x Interest rate change			0.021 (0.052)	0.022 (0.058)	0.004 (0.058)
Observations	129,027	100,263	129,027	100,263	100,263
R-squared	0.149	0.151	0.169	0.170	0.179
Estimation	OLS	2SLS	OLS	2SLS	2SLS
Firm controls	no	no	no	no	yes
Time sector FE	yes	yes	yes	yes	yes
Time clustering	yes	yes	yes	yes	yes

Notes: In Columns (1) and (2), we report regression results from estimating the following specification: $\Delta \log k_{it+1} = \alpha_i + \alpha_{jt} + \beta \Delta \log q_{it} + \epsilon_{it}$ where α_i is a firm fixed effect, α_{jt} is a sector-time fixed effect, $\Delta \log q_{it}$ is real RE price growth in MSA k at time t , using OLS and 2SLS respectively. In Columns (3) and (4), we report regression results from estimating the following specification: $\Delta \log k_{it+1} = \alpha_i + \alpha_{jt} + \beta_1 \Delta \log q_{it} + \beta_2 mpk_{it-1} \Delta \log q_{it} + \gamma_2 mpk_{it-1} \Delta r_t + \Gamma' Z_{it-1} + \epsilon_{it}$ where α_i is a firm fixed effect, α_{jt} is a sector-time fixed effect, $\Delta \log q_{it}$ is real RE price growth in MSA k at time t , mpk_{it-1} is the log of the MPK of firm i at time $t-1$, Δr_t is the change in real mortgage rates at time t , and Z_{it-1} is a vector containing mpk_{it-1} , using OLS and 2SLS, respectively. In column (5), we re-estimate the 2SLS regression in column (4) with an enlarged set of firm-level control variables, Z_{it-1} , containing MPK, sales growth, size, current assets as a share of total assets, total debt as a share of total assets, and the interaction of MPK with lagged real GDP growth. All explanatory variables (notably MPK) except RE price growth and the interest rate change are standardized over the entire sample. Standard errors are two-way clustered by firms and quarter. Standard errors in columns (2), (4), and (5) are bootstrapped based on 500 simulations. *** Significant at the 1 percent level. ** Significant at the 5 percent level.

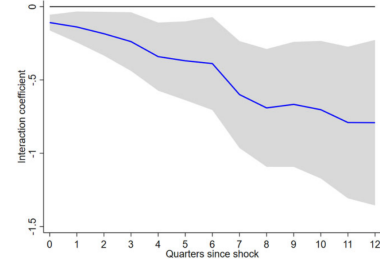


FIGURE 8
Dynamics of differential response of investment to real estate price changes

Notes: The figure displays the regression coefficients β_{2h} (the solid line) obtained when estimating the following specification: $\log k_{it+h} - \log k_{it-1} = \alpha_{ih} + \alpha_{jh} + \beta_{1h} \Delta \log q_{it} + \beta_{2h} mpk_{it-1} \Delta \log q_{it} + \gamma_{1h} \Delta r_t + \gamma_{2h} mpk_{it-1} \Delta r_t + \Gamma_h' Z_{it-1} + \epsilon_{ih}$ for $h = 0, 1, 2, \dots, 12$ and Z_{it-1} is a vector containing mpk_{it-1} . Regression is estimated using 2SLS with the interaction between the real estate supply elasticity and the real mortgage rate as instrument for real estate price growth in the first stage. The shaded area depicts the 90% confidence interval. All variables are as defined in Table 2. Standard errors are two-way clustered by firms and quarter.

- The cumulative interaction effect remains negative and statistically significant up to twelve quarters.
- This matters because the theory is about persistent reallocation, not only impact responses. The figure confirms that the differential crowding-out of high-MPK firms is persistent.

6 Short summary

- **Conclusion:** Lower interest rates can reduce output when they reallocate scarce capital from more productive, financially constrained firms toward less productive firms.
- **Intuition:** The price of capital rises in general equilibrium. That extra price pressure benefits some firms but crowds out the most productive constrained users of capital.
- **Empirical lesson:** To understand the macro effects of low rates, one must track how asset prices shift the allocation of investment across firms, not only how they change aggregate borrowing conditions.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding:** The sign of the output effect of lower rates is theoretically ambiguous once one allows for heterogeneous productivity, borrowing constraints, and an imperfectly elastic supply of capital.
- **Magnitude:** In the calibration, the long-run reallocation effect is large enough to dominate the positive capital-supply effect, producing an output decline of around 5.2%.
- **Credibility:** The theory is transparent, the dynamic extension shows how the mechanism interacts with balance-sheet effects, and the empirical analysis directly tests the cross-firm prediction of the model.

- **Mechanism evidence:** Lower-MPK firms react more strongly to interest-rate-induced local real-estate price increases than higher-MPK firms do.
- **Policy implication:** The competitive equilibrium is constrained inefficient because low-productivity investors do not internalize the crowding-out effect they impose on higher-productivity constrained firms.
- **Additional discipline:** The result is robust to richer technologies, heterogeneous wealth, dynamic wealth accumulation, and alternative monetary-shock measures.

7.2 Economic intuition

- Lower rates do two things at once: they relax financing conditions and they bid up the price of scarce collateralizable assets.
- The first effect helps firms borrow; the second effect reallocates scarce assets toward firms that should not necessarily be the marginal users from a social standpoint.
- In the short run, capital gains help productive borrowers. In the long run, however, the persistent price distortion matters more than the temporary balance-sheet windfall.

7.3 Takeaway

This is a clean paper about how monetary and financial conditions affect *allocation*. The usual story says that low rates stimulate investment by easing finance. This paper shows that, once scarce assets have an endogenous price, low rates may instead move capital toward lower-return users and depress productivity. The main lesson is that the macroeconomic consequences of low rates depend not only on how much firms invest, but on which firms get to invest.